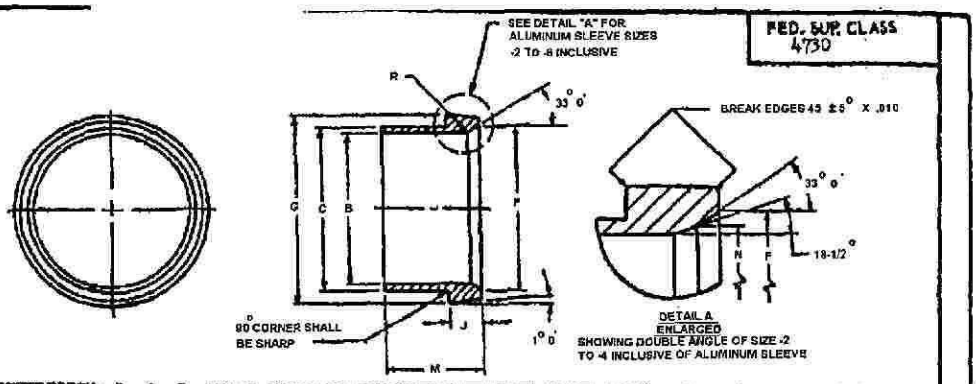
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4730



(a) CONCENTRICITY: B, C, F, AND G SHALL BE CONCENTRIC TO EACH OTHER WITHIN .005 TOTAL INDICATOR READING FOR TUBE SIZES UP TO AND INCLUDING 3/4 INCH, AND .010 TOTAL INDICATOR READING FOR SIZES 1 INCH AND ABOVE.

(b) OPTIONAL METHOD OF MANUFACTURE: STEEL SLEEVES MAY BE FABRICATED FROM MIL-T-5066 (G1025) SEAMLESS STEEL TUBING. THE 1° ANGLE ON THE SHOULDER OF THE SLEEVE IS NOT REQUIRED WHEN FABRICATED FROM SEAMLESS STEEL TUBING.

WARPING OF SLEEVES FROM ANY CAUSE SHALL NOT BE CAUSE FOR REJECTION IF THEY FIT THE MINIMUM DIMENSIONS OF THE AN818 NUT WHEN ASSEMBLED ON A LENGTH OF TUBING.

Ⓒ MATERIAL: STEEL, MIL-S-6758 (4130), MIL-S-5626 (4140), QQ-S-637 (G1137, G1141); CORROSION RESISTANT STEEL (CRES), QQ-S-763, CLASS 304 OR 316; ALUMINUM ALLOY, QQ-A-225/4 (2014), QQ-A-225/6 (2024); BRASS, QQ-B-626, COMPOSITION 21 OR 22. SEE PROCUREMENT SPECIFICATION.

FINISH: STEEL, CADMIUM PLATE (QQ-P-416, TYPE II, CLASS 2, BLACK). CRES, PASSIVATE (MIL-S-5002). ALUMINUM ALLOY, ANODIC TREATMENT (MIL-A-8625, TYPE II).

BREAK ALL SHARP EDGES AND REMOVE ALL HANGING BURRS AND SLIVERS.

DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES: DECIMALS ±.005, ANGLES ±1/2°.

NO CODE LETTER IN PART NUMBER FOR STEEL. ADD B AFTER DASH NUMBER FOR BRASS. ADD D AFTER DASH NUMBER FOR ALUMINUM ALLOY. ADD J AFTER DASH NUMBER FOR CRES, TYPE 304. ADD K AFTER DASH NUMBER NUMBER FOR CRES, TYPE 316.

Ⓒ ITEMS CODED C ON PREVIOUS ISSUES OF THIS STANDARD ARE NO LONGER TO BE FURNISHED. PROCURING ACTIVITIES MUST SPECIFY APPLICABLE CRES MATERIAL CODE AS PART OF THE STANDARD PART NUMBER.

EXAMPLE OF PART NUMBERS:

MS20819-5 – SLEEVE, 5/16 TUBING OD, STEEL. MS20819-5B – SLEEVE, 5/16 TUBING OD, BRASS.
MS20819-8D – SLEEVE, 1/2 TUBING OD, ALUMINUM ALLOY. MS20819-8J – SLEEVE, 1/2 TUBING OD, CRES (304).
MS20819-8K – SLEEVE, 1/2 TUBING OD, CRES (316).

FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

ALUMINUM ALLOY SLEEVES (CODE D) ARE INACTIVE FOR BOTH NEW DESIGN AND REPLACEMENT PARTS. FOR REPLACEMENT SLEEVES SUBSTITUTE CADMIUM PLATED STEEL (NO CODE) OR CRES (CODE J OR K).

DASH NO.	TUBING OD	(a) B +.003 −.000 DIA	(a) C +.000 −.003 DIA	(a) F DIA	(a) G +.000 −.003 DIA	J ±.016	H ±.016	N DIA AL. ALLOY	R ±.016 RAD STEEL	R ±.016 RAD AL ALLOY	WEIGHT MAX (LBS.) STEEL	WEIGHT MAX (LBS.) AL ALLOY	WEIGHT MAX (LBS.) CRES	WEIGHT MAX (LBS.) BRASS
2	1/8	.130	.172	.205	.267	.125	.344	.178						
3	3/16	.193	.234	.267	.329	.141	.406	.240	.031					
4	1/4	.255	.297	.315	.383			.288						
5	5/16	.318	.366	.375	.445	.156	.438	.348						
6	3/8	.380	.432	.441	.502	.172	.500	.414	.047					
8	1/2	.505	.562	.589	.682	.219	.562	– –	.062	.062				
10	5/8	.631	.690	.705	.797	.234	.656	– –						
12	3/4	.756	.826	.880	.972	.266	.688	– –	.078	.078				
16	1	1.006	1.081	1.130	1.222	.281	.781	– –						
20	1-1/4	1.260	1.339	1.412	1.534	.312	.906	– –	.094	.094				
24	1-1/2	1.510	1.609	1.630	1.784	.344	1.125	– –						
28	1-3/4	1.762	1.882	1.945	2.159	.375		– –						
32	2	2.014	2.159	2.195	2.409	.406	1.188	– –	.109	.109				
40	2-1/2	2.516	2.659	2.695	2.911	.469	1.312	– –						
48	3	3.016	3.172	3.195	3.411	.531	1.375	– –						

Ⓒ REDRAWN

Ⓒ denotes changes

P.A. Air Force-99 Other Cust Navy – AS	Interchangeability Interface	TITLE SLEEVE, FLARED TUBE FITTING	MILITARY STANDARD MS20819
Procurement Specifications MIL-F-5509		SUPERSEDES AN819	PAGE 1 OF 1

DD FORM 1 MAY 73 672 (Coordinated) PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

User activities: Army – ME,AV,CL,MI Navy – AS Air Force – 99

Activities: Army – ME,AV Navy – AS Air Force – 99 DLA – GS

Rev.

This standard is approved for use by all Departments and Agencies of the Dept. of Defense. Selection for all new engineering and design applications and for repetitive use shall be made from this document when applicable.

APPROVED 26 AUG 57 REVISED Ⓒ 21 MAR 79

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